

MA-105 Tutorial-9

Daksh Maahor

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1. Evaluate the following integral over the surface T , where T is given by the closed triangular region defined by the points $(0, 0)$, $(1, 3)$ and $(2, 2)$.

$$\int \int_T e^{x-y} dx dy$$

2. Compute :

$$\int \int_D y^3 (x^2 + y^2)^{-\frac{3}{2}} dx dy$$

over the surface defined as $D = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 \leq 1, \frac{1}{2} \leq y \leq 1\}$

3. (a) Parametrize the curve of intersection of

$$Ax^2 + By^2 + Cz^2 = \alpha$$

and

$$A^2x^2 + B^2y^2 + C^2z^2 = \beta$$

lying in the positive octant by supplementing it with

$$x^2 + y^2 + z^2 = t$$

(b) Compute $\frac{\partial(\alpha, \beta, t)}{\partial(x, y, z)}$

4. Parametrize the curve of intersection of $x^2 + y^2 + z^2 = 4a^2$ and $(x - a)^2 + y^2 = a^2$. Set up the integral to calculate its arc length and compute it.
5. Evaluate:

$$\int_0^4 \int_0^1 \int_{2y}^2 \frac{4 \cos(x^2)}{2\sqrt{z}} dx dy dz$$